LUMBINI ENGINEERING COLLEGE

Rupandehi, Nepal

2000

Pre Board

Level: BE Computer Engineering 3^{rd} Year | 7^{th} Semester

Subject: Data Science and Analytics

Time: 3 hrs

FM: 100

PM: 45

Candidates are required to give their answers in their own words as far as practicable.
The figure in the margin indicates full marks.

Attempt all the questions

1. (a) Why do we require data transformation? Explain by giving realistic examples. 7

   (b) Differentiating between numeric and categorical variables. A numeric variable has the following values. Standardize the values. 8
   46.93, 138.07, 64.77, 51.46, 15.38, 15.37, 11.32, 223.21, 120.23, 19.85, 19.80, 12.39, 55.79, 86.31, 3.99, 101.51, 30.71, 94.28, 102.39, 18.04

2. (a) Explain Pearson correlation and how it measures the relationship between two numeric variables by considering the example. 7

   (b) A company wants to determine if the preference for a new product (*Like* or *Dislike*) is independent of the gender of the consumer. A survey was conducted among 200 individuals, and the following contingency table was obtained: 8

Gender	Like	Dislike	Total
Male	60	40	100
Female	80	20	100
Total	140	60	200

Question:

i. State the **Null Hypothesis** (H_0) and the **Alternative Hypothesis** (H_a) for this study.

ii. Calculate the **Chi-squared** (χ^2) statistic. Show the calculation for the expected frequencies (E) for each cell.
Note: The formula for the test statistic is $\chi^2 = \sum \frac{(O-E)^2}{E}$

iii. Given that the critical value for $\alpha = 0.05$ with $df = 1$ is 3.84, determine whether there is a significant relationship between gender and product preference. State your conclusion clearly.

3. (a) Define Multi-collinearity and explain how Variance Inflation Factors (VIF) are used to detect it. Discuss the trade-offs between selecting a linear regression model versus a non-parametric model (like Nadaraya-Watson) for real-world business datasets. 8

(b) You are given the following two-dimensional data set consisting of 6 observations: 7

(2,1),(3,5),(4,3),(5,6),(6,7),(7,8)

Using the Principal Component Analysis (PCA) algorithm, compute the first principal component of the dataset

4. (a) Compute the R-Squared (R^2) for a regression model to measure how well it explains the variation in the dependent variable given the following data: 8

Hours Studied (X)	1	2	3	4
Exam Score (Y)	50	60	70	80

The fitted regression model is: $\hat{Y} = 40 + 10X$.

(b) Discuss the challenges of K-means clustering when dealing with mixed-type data. What distance metrics are appropriate for datasets containing both numeric and categorical variables? 7

5. (a) You are given the following dataset of customers, where the target variable is whether the customer purchased a product (*Yes* or *No*). Using the **CART Algorithm**, construct a binary classification tree up to **two levels deep** for predicting whether a customer will purchase the product. 8

Customer	Age	Salary	Purchased
1	Young	High	No
2	Young	Medium	Yes
3	Middle	High	Yes
4	Senior	Low	No
5	Senior	Medium	Yes
6	Middle	Low	Yes

(b) Explain the working of logistic regression. Compare and contrast Decision Trees with Logistic Regression for a binary classification task. 7

6. (a) What do you mean by Causation explain with suitable example. Differentiate between ARMA and ARIMA model based on components of models. 8

(b) What is Granger Causality? Explain Causal Directed Acyclic Graph in Details. 7

7. Write short note (any two) [5 * 2 = 10]

(a) Power laws vs Zipf's law

(b) Variable Importance: permutation tests, partial dependence plots

(c) R^2 vs. Adjusted R^2 Score